Aussie Primary Academy

States of Matter — Year 5 Science

Australian Curriculum v9.0 | AC9S5U04 | Chemical Sciences

LEARNING INTENTION

I can explain the properties of solids, liquids and gases using particle models.

SUCCESS CRITERIA

☐ I can describe the properties of each state of matter
☐ I can explain particle arrangement and movement
☐ I can identify changes of state with real examples

KEY CONCEPTS

Solid: particles tightly packed, fixed shape. **Liquid:** particles close but can move, takes shape of container. **Gas:** particles spread far apart, fill any space.

WORKED EXAMPLE

Ice (solid) melts to water (liquid) then evaporates to steam (gas). Each state change is caused by adding or removing **heat energy**.

PART A — WARM UP

1. Classify each: milk, stone, steam, honey, oxygen, brick
2. What happens to particle movement when a solid is heated?
3. Why does a gas fill its entire container?
4. What change of state occurs when water freezes?

PART B — CORE SKILLS

5. Draw or label particle diagrams for a solid, liquid and gas in your working space.
6. Name the change of state: liquid → gas: ____; gas → liquid: ____; solid → liquid: ____
7. Explain why a liquid takes the shape of its container but a solid does not, using particle theory.
8. Give a real-world example of each change of state from Question 6.

STATES OF MATTER HELPER

Changes of state: melting (solid → liquid), freezing (liquid → solid), evaporation (liquid → gas), condensation (gas → liquid). All are caused by changes in **heat energy** — no new matter is created or destroyed.

WORKING SPACE — PARTICLE DIAGRAMS

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PART C — CHALLENGE

9. Real-World Context: When you hang wet washing outside on a hot day, the water disappears into the air. Explain this process using the terms: particles, heat energy, evaporation and water vapour.

10. Reasoning: A friend says "steam is just water that has disappeared." Using your knowledge of particle theory and states of matter, explain why this is not correct.

REFLECTION

Today I learned…

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I CAN CHECKLIST

☐ I can describe properties of solids, liquids and gases

☐ I can explain particle arrangement

☐ I can identify and name changes of state

TEACHER FEEDBACK

Strength: ______________________

Next Step: ______________________

PARENT TIP

Boil water together and observe steam forming — talk about what's happening to the particles.

TEACHER TIP

Use jelly beans as "particles" that students physically move — tight cluster (solid), loose (liquid), spread out (gas).

ANSWER KEY (FOR PARENT / TEACHER USE)

Q1: Solid: stone, brick. Liquid: milk, honey. Gas: steam, oxygen
Q2: Particles move faster and vibrate more
Q3: Particles in gas are spread far apart with no fixed shape
Q4: Freezing (liquid → solid)
Q5: Check diagrams — solid: tight grid; liquid: close but random; gas: widely spread
Q6: Liquid → gas: evaporation; gas → liquid: condensation; solid → liquid: melting
Q7: Liquid particles can slide past each other; solid particles are locked in place by strong forces
Q8: Evaporation: wet clothes drying; condensation: dew on glass; melting: ice cream in sun
Q9: Heat energy from the sun causes water particles to gain energy, move faster and escape into the air as water vapour (gas) through evaporation
Q10: Steam is water vapour — the particles still exist, just spread out as a gas. Matter is not destroyed, only changes state